Kalshi and the Institutionalization of Macro Prediction Markets

What a validated forecasting instrument does, and does not, earn the right to become

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ABSTRACT

Diercks, Katz and Wright (2026) showed that Kalshi's macro prediction markets forecast about as well as the professionals. On the policy rate they are statistically indistinguishable from the New York Fed's survey and from fed funds futures, and on headline CPI they beat the Bloomberg consensus. This project takes that result as settled and does not try to forecast anything better. It asks the question the paper sets aside: now that the instrument is accurate, should a public institution build it into how it measures expectations, communicates, and sets policy, and if so, how far, and with what protections.

The view defended here is that accuracy earns the instrument a place in the panel and little more. A market can be a good forecaster and a poor thing to govern by. The moment the Fed leans on a Kalshi number, three quiet problems become central: a thin market can be moved by a single legal bet placed to shape a headline; a central bank citing a market that bets on its own decisions risks a feedback loop in which the signal is no longer independent of the policymaker reading it; and the "public" being priced is a KYC'd, retail slice, not the public as a whole. The recommendation is deliberately modest: a safeguarded, clearly labeled data product that supplements the existing tools rather than replacing them, gated on a liquidity floor, pre-meeting concentration monitoring, and honest distributional framing.

One caveat belongs up front. There is a working, tested Python pipeline that re-implements the paper's ladder-to-density method and adds the tooling a governance argument needs, but the empirical sections currently run on synthetic data. Every notebook ships a generator so it runs offline, and every place a real Kalshi number belongs is marked as a placeholder. Where the pipeline is said to "replicate Figure 1," it reproduces the shape on simulated ladders; it has not been re-plotted on an independent pull. That pull is the next task, not a finished one.

1 Why accuracy is the beginning, not the end

The FEDS paper answered a clean empirical question, and answered it well: Kalshi is accurate. Fed funds forecasts indistinguishable from the New York Fed's Survey of Market Expectations and from fed funds futures; headline CPI beating the Bloomberg consensus; a model-free method for turning a ladder of "exceeds strike X" contracts into a full risk-neutral distribution; and event-study evidence that news moves the variance and skew of that distribution, not only its mean. None of that is in dispute here.

But a good forecast and a good thing to govern by are not the same object. Once an official process leans on a market-implied number, its accuracy stops being a private matter for traders and becomes a public one, and four things turn from footnotes into first-order concerns.

- Signal integrity becomes a public good. If the Fed cites a price, moving that price stops being a bet and becomes a way to move public policy.
- Legitimacy is not accuracy. A central bank pointing at a market where the public bets on the central bank's own decisions must justify why a betting line is evidence for a public decision.
- Transparency cuts both ways. A published distribution is more honest than a proprietary consensus only if it is labeled for what it is, what traders are pricing rather than what the Fed expects.
- The "public" on Kalshi is a KYC'd, retail-tilted, self-selected venue. Reading its price as "the market," let alone "the people," is a choice made explicitly rather than by default.

None of these is answered by "the forecasts are accurate." They are the substance of what follows.

2 The replication, and what it proves right now

The replication exists so the new arguments rest on a result reproduced here rather than only cited. `notebooks/01_figure1_replication.ipynb` takes a synthetic ladder that noisily converges to each meeting's realized rate, pushes each day through `ladder_to_pdf`, and aggregates error by days-before-meeting exactly as Figure 1 does. It reproduces the shape the paper reports, with error shrinking into the meeting and the mode near zero by the day before. That confirms the machinery, not the data: there is no real Kalshi pull in it yet. `candlesticks_to_daily_ladder` is the one bridge left once a pull exists, and crossing it is the top of the work order.

One addition strengthens the method. The ladder-to-density step is grounded in the contract taxonomy of Snowberg, Wolfers and Zitzewitz (2012): a Kalshi "exceeds strike X" ladder is a family of winner-take-all contracts, so its price schedule across strikes is the risk-neutral CDF, and differencing recovers the density. That gives Section 3's method a textbook footing independent of the paper's own derivation, and is why the same code runs unchanged on CPI, GDP and recession ladders. The Table 3 comparison (MAE/RMSE against Bloomberg and futures, Diebold-Mariano tests) is coded and tested in `forecast_eval.py` and waits on the same pull plus the NY Fed survey PDFs. If the proprietary Bloomberg consensus cannot be obtained, a public aggregator will be substituted and the substitution flagged, since a silent swap would make this a "reproduction" in name only.

3 Manipulation: the deep market is safe, the thin ones are not

The paper gives manipulation a single sentence, one citation to Hanson and Oprea (2009). For a thesis of “accurate enough to use,” that is thin on the first question a policymaker asks: can this number be pushed, and where? The claim here is narrow and defensible. The most-watched series is very hard to move; the thin series are not, and it is the thin series a motivated actor would target.

The $7 million per-market cap the paper presents as a sign of maturity reads differently from the other side. Against fed funds volumes running to roughly a hundred million contracts around a meeting, $7M is a rounding error and the self-correcting story holds. Against a GDP or recession-probability book with a two-year history, $7M could be a large share of everything outstanding, so the same cap is either binding but evadable or simply large relative to the book. The Hanson and Oprea result is also an equilibrium statement, that markets stay informative on average because a manipulator effectively pays informed traders to correct him. It does not promise that any single snapshot, in a thin market, in the 24 to 48 hours before a decision, is manipulation-proof, because the correcting money may not arrive in time. The fuller record agrees: in liquid markets manipulation is typically transitory and self-correcting, and the documented failure mode is thin markets without enough noise traders, which is precisely the GDP and recession series.

The scenario a staffer should weigh is not the illegal one. Insider trading is already unlawful, a legal problem rather than a market-design one. The harder case is legal: a large, permitted, directional bet in a thin Fed-adjacent market whose point is the “markets expect X” headline it generates, not the profit, the headline the press repeats and a citing central bank then treats as evidence. A citation policy has to account for it.

4 The adoption ladder, and where Kalshi already sits

Formal use is a ladder, and the risks rise with each rung: (a) an internal staff monitoring tool; (b) a public staff-research citation; (c) a communications-level citation, a governor saying “markets are pricing X% odds” from the podium; (d) a standing public data feed, as the NY Fed publishes SOFR and repo rates. The paper is itself evidence the climb has begun. It is a Board FEDS paper out of the Division of Monetary Affairs, and it proposes, subject to approval, a public distributional dashboard at EconFutures.com, in substance a monitoring dashboard. So Kalshi already sits at roughly rung (a) to (b). The honest task is to measure the next rung against how the Fed already treats OIS-implied paths, futures-implied probabilities and TIPS breakevens, the real benchmark rather than “nothing,” and to leave the specific citations as items to verify rather than to invent.

5 Regulated versus unregulated: does the CFTC stamp buy better signal?

Polymarket appears once in the paper, set aside as a “legal gray area” with fewer contracts and thinner liquidity. The paper leans on Kalshi’s regulated status but never tests whether it buys a better signal or only a cleaner story. This project runs the head-to-head on the same pipeline: Polymarket lists a Fed decision as sibling yes/no markets, so `fed_ladder_from_markets` assembles them into a ladder that feeds the identical `ladder_to_pdf` path. The expected result is worth stating in advance, because being honest about the confound is the point. The two platforms differ in user base, contract design, settlement currency, position limits and listing history at once, so a clean causal “CFTC regulation to better signal” claim is not identified. The modest expectation: regulation probably buys tighter, less stale markets and a real manipulation backstop, but a decisive forecast-accuracy edge will be hard to establish, and that is itself the finding the one-sentence dismissal never earned.

6 What you do with a distribution, and the one thing you should not

The question is not “should we cite it” but “once we trust it, what can we do with a distribution that a survey point or a single futures probability cannot.” The discipline is one line: the mean is largely redundant, the distribution is not. The average of a Kalshi series mostly re-tells what the futures strip, OIS path and breakevens already say. The new object is the full shape, the variance and skew and tails, for horizons and variables where no option-implied distribution exists. The paper’s own event study is the load-bearing fact: the higher moments respond to policy events, so they carry information a mean-only reader discards. `policy_signals.py` makes this concrete with one number, `opposite_tail_prob`, the probability of a move against the market’s expected direction. A futures quote can say “the market expects roughly a hold” and nothing more; it cannot say the market at the same time prices a 30% cut and a 30% hike around that flat expectation. That omitted probability is the exact, computable cost of collapsing a distribution to its mean.

The channels where the distribution earns its keep: gap-detection between the quarterly SEP dots and the market-implied path on the roughly eighty silent days between projections; a real-time read on the asymmetry a risk-management framing consumes; the recession-probability series as a traded, continuously updated cross-check on GDPNow and the Sahm rule; and the uncertainty of the rate path, not just the path, as an input to Treasury debt management, lower risk because issuance is not what the market is betting on. One channel the project recommends against: a market-implied recession probability should not be wired into a statutory fiscal trigger, which must key off a tamper-proof official statistic rather than a price a thin market can move, and whose incentive to be moved would rise the moment fiscal consequences were attached. Naming a channel that should not be built is the discipline that makes the rest credible.

The finding with the strongest claim to originality runs against the intuition: manipulation risk and reflexivity risk point in opposite directions across the menu of series. Signals are safest where the market prices something the policymaker does not control (CPI, GDP, recession) and most fraught where it prices the policymaker’s own choice (the fed funds decision). So the deepest, most manipulation-resistant series is the very one where reflexivity bites hardest, while the thin real-economy series, the ones most exposed to manipulation, are where the signal is genuinely exogenous. Liquidity says fed funds first; exogeneity says fed funds last. Reconciling the two is the real result: the safe order is real-economy-first and policy-rate-last, the mirror image of what liquidity alone would suggest.

7 The recommendation, stated plainly

A Kalshi distributional dashboard should supplement the existing tools, the survey, Bloomberg, the SPF and futures, and should not replace them. That is the correct relationship, not timidity, because the paper’s own results show Kalshi ties or modestly beats the incumbents rather than dominating them. No series should be citable in official communications until three safeguards are in place.

1. A minimum liquidity floor. Below a set open-interest or trailing-volume threshold, a price is not a citable expectations measure, calibrated so a $7M position cannot move the implied probability by more than a few points.
2. Pre-meeting concentration monitoring. Flag any FOMC-week move driven by a handful of large accounts before calling it “market expectations,” using the large-trader reporting Kalshi owes as a CFTC-regulated exchange and Polymarket does not.
3. Distributional framing with a disclaimer. Show mean, median, mode, variance and skew together, labeled as a retail-market-derived measure, “what traders are pricing,” not “what the Fed expects.”

A fourth follows from the reflexivity finding: sequence by exogeneity, not just liquidity. Lead with the real-economy distributions and treat the fed funds distribution as a labeled cross-check rather than the headline. The concrete first step is deliberately small: a quarterly appendix chart in the Monetary Policy Report, or a NY Fed markets-desk-style webpage carrying only series that clear the floor. That is more conservative than “cite it in the FOMC statement,” and on purpose, because rung (c) is where the “central bank points at a market betting on its own decisions” problem is sharpest.

8 What holds the argument up, in code

There is a tested `src/` package behind the prose, installed via `pyproject.toml`, linted, and run in CI on Python 3.11 and 3.12 including offline execution of all three notebooks.

Module	What it does	Grounds
`kalshi_utils.py`	Section 3 method: ladder to risk-neutral pdf to mean, median, mode, variance, skew; `candlesticks_to_daily_ladder` bridges the API	Replication
`ladder_validation.py`	Explicit no-arbitrage screen (prices in [0,1], non-increasing in strike), a gate before the silently-clipping `ladder_to_pdf`	Replication, Manipulation
`policy_signals.py`	Reads the ladder on the 25bp FOMC grid to a cut/hold/hike distribution, modal and expected move, dispersion, and `opposite_tail_prob`	Implementation
`forecast_eval.py`	RMSE, MAE, bias, Diebold-Mariano, the machinery behind Table 3	Replication

Module	What it does	Grounds
`calibration.py`	Brier score, Murphy decomposition, reliability table, testing the probabilities as probabilities	Replication
`event_study.py`	How variance and skew jump around each event, Section 7 turned into a table	Replication, Implementation
`manipulation.py`	Prototype cost-to-move: dollars needed to shift the distribution, scaling with market depth	Manipulation
`kalshi_api.py` / `polymarket_api.py`	Thin read-only market-data clients; both platforms' endpoints are public	Replication, Polymarket

Two are worth singling out. `opposite_tail_prob` is the number that carries the implementation argument, the scalar a point forecast cannot report. And `ladder_validation` exists because `ladder_to_pdf` is defensive by design: it clips negative masses and renormalizes, right for production but capable of turning a malformed pull into a plausible-looking distribution silently. When the project is about official reliance, that is exactly the failure to close.

9 Where this is real, and where it is scaffolding

Real and done: the full `src/` pipeline, tested, linted, green in CI; all seven section drafts as complete arguments; the three notebooks running end to end, offline, on synthetic data; and the supporting literature, Sumner (2018) on targeting the forecast and Snowberg, Wolfers and Zitzewitz (2012) on manipulation and contract taxonomy, woven into the prose.

Scaffolding, argument committed but numbers not: there is no real Kalshi pull yet, so Figure 1 and Table 3 are reproduced in shape on synthetic ladders, the single biggest gap. The liquidity comparison, likely the most persuasive original evidence the project can offer, runs on synthetic volumes today. The cost-to-move figures, the $7M-as-share-of-open-interest numbers, and the weak-form-efficiency screen wait on real order-book data. Every specific institutional citation, which FOMC minutes reference OIS, whether the CFTC has acted against a DCM prediction market, whether Treasury references rate distributions, is marked as needing verification and not asserted until checked. `TODO.md` tracks every one against the pull that fills it.

10 Bottom line

The FEDS paper did the hard empirical work convincingly, and none of this is possible without it. What it leaves open is the governance layer: the manipulation surface of the thin series, the reflexivity problem that runs opposite to liquidity, the one channel that should not be built, and a concrete safeguarded first step short of the riskiest form of official citation. Accuracy gets the instrument into the room. Whether it should shape a public decision is a different question, and the answer is yes, carefully, labeled for what it is, and never as the thing policy is calibrated to.